

Удивительный мир VSR метрик (pt. 2)



Таксономия метрик

1. Level

temporal

frame-level

subjective

2. Computation logic

pixel-based

DL

3. Evaluation intent

fidelity

perceptual
quality

realism

How much does the output
look like the pixel-perfect
Ground Truth?

How "sharp" or "pleasing"
does it look to a human?

Does the added detail
look like a real video or AI
noise?

↑ MUSIQ (Multi-scale Image Quality Transformer)

frame-level

DL

perceptual
quality

+ no-reference

How does it work?

By Google Research.

It was designed to solve a fundamental flaw in how older deep-learning metrics (like those based on CNNs) judge images: MUSIQ can look at a 4K upscaled image in its native resolution.

- 1) **Multi-Scale Representation**: MUSIQ takes the **original resolution** image and several downsampled versions (high-res version captures Technical Quality, low-res versions capture Aesthetic Quality).
- 2) **Patch-Based Processing**: the model breaks these multi-scale images into fixed-size patches.
- 3) **Three Specialized Embeddings**: to help the Transformer understand where each patch came from, MUSIQ uses three layers of "labels": a) **spatial embedding** (hash-based 2D grid); b) **scale embedding** (which level of the pyramid (which resolution) the patch belongs to); c) **patch embedding** (actual pixel content)

It currently holds some of the *highest correlations* with human **Mean Opinion Scores (MOS)** on major benchmarks like KonIQ-10k and PaQ-2-PiQ.

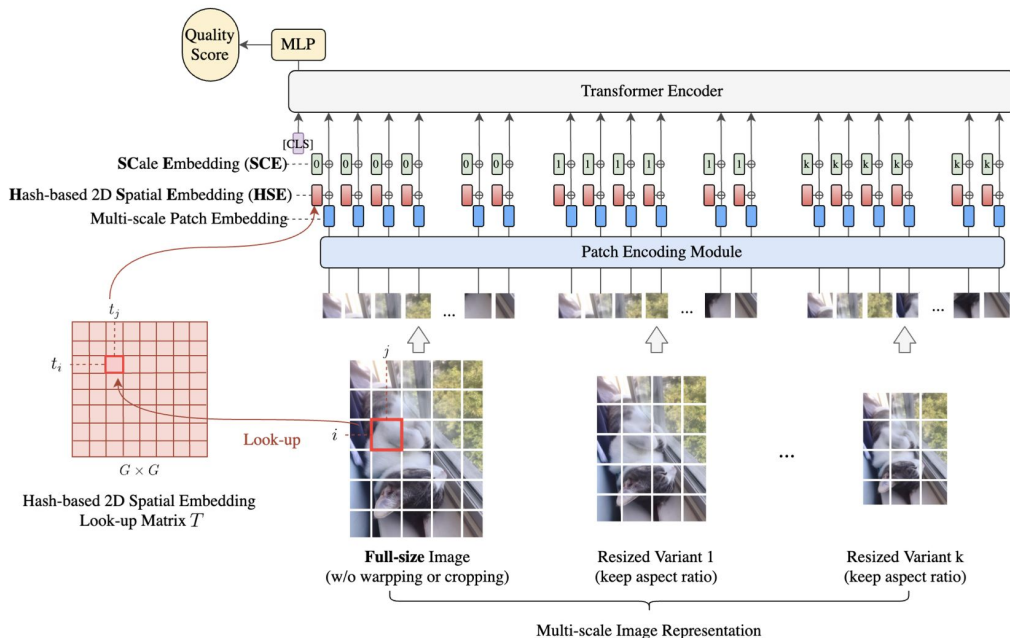
↑ MUSIQ (Multi-scale Image Quality Transformer)

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↑ MANIQA (Multi-dimension Attention Network for No-Reference Image Quality Assessment)

frame-level

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How does it work?

Multi-Dimensional Attention

It looks at an image through two different lenses at the same time: **Channel Attention** and **Spatial Attention**. It was specifically designed to identifying the "plasticity" or "hallucinated" textures common in AI-upscaled images.

- 1) **Feature Extractor (ViT)**: break the image into 8×8 patches.
- 2) **TAB (Transposed Attention Block)**: this block applies **Channel Attention**. Instead of looking at where the pixels are, it looks at the different "filters" (channels) of the data.
- 3) **SSTB (Scale Swin Transformer Block)**: this block applies **Spatial Attention**. It checks for local consistency.
- 4) **Dual Branch Prediction**: it uses a unique structure to calculate the score:
 - a) **Scoring Branch**: Predicts a quality score for every single 8×8 patch;
 - b) **Weighting Branch**: Predicts how much "importance" each patch should have

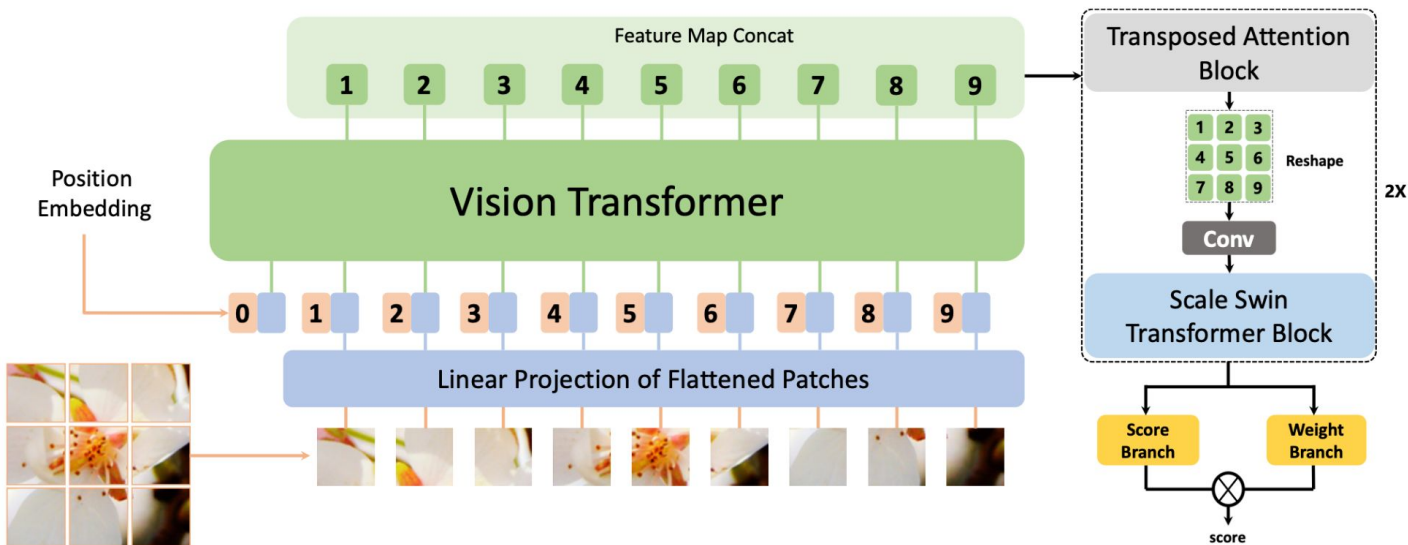
↑ MANIQA (Multi-dimension Attention Network for No-Reference Image Quality Assessment)

frame-level

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↓ NIQE (Natural Image Quality Evaluator)

frame-level

pixel-based

realism

+ no-reference

How does it work?

The Philosophy: Natural Scene Statistics (NSS)

NIQE is based on the idea that all "natural" (unprocessed, high-quality) images share certain mathematical properties.

- 1) **Preprocessing**: the image is normalized using **Mean Subtracted Contrast Normalization (MSCN)**. This removes the "content" of the image (the objects) and leaves behind the "textures" and "noise."
In a natural image, these MSCN coefficients always follow a perfect Gaussian (bell-curve) distribution.
- 2) **Feature Extraction**: divides the image into patches and extracts a set of "quality-aware" features from these coefficients.
- 3) **Multivariate Gaussian Model Comparison**: NIQE has a pre-calculated **Multivariate Gaussian (MVG)** model made from thousands of pristine natural images. It creates a similar **MVG** model for your specific test image. It calculates the distance between these two models.

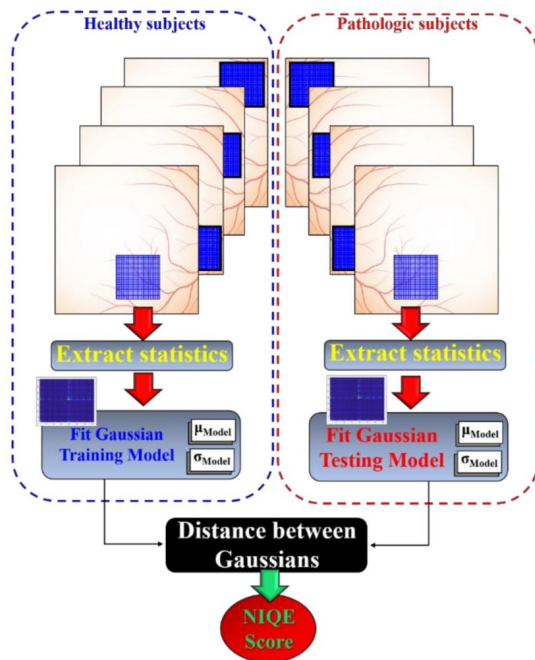
↓ NIQE (Natural Image Quality Evaluator)

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↓ DISTS (Deep Image Structure and Texture Similarity)

frame-level

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How does it work?

"Texture vs. Structure" trade-off

$$s(\tilde{x}, \tilde{y}) = \frac{2\sigma_{xy} + c_2}{\sigma_x^2 + \sigma_y^2 + c_2}$$

$$l(\tilde{x}, \tilde{y}) = \frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1}$$

DISTS uses a pre-trained **VGG-16** network as a feature extractor. However, it doesn't just compare the raw features (like LPIPS does); it applies a mathematical logic inspired by SSIM to those deep features.

- 1) **Feature Extraction**: the Reference (x) and the Distorted (y) images are passed through VGG-16. DISTS extracts feature maps from five specific convolution layers
- 2) **Two Key Components**: DISTS calculates two separate similarity indices:
 - a) **Structure Similarity (s)**: This uses the **correlation** between feature maps. It ensures that the "shapes" and "content" of the images align.
 - b) **Texture Similarity (l)**: This compares the **global statistics** (mean and variance) of the feature maps.
- 3) **Weighted Summation**: The final DISTS score is a weighted sum of these similarities across all layers. These weights were specifically optimized to match human ratings and to be **invariant to texture resampling**.

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